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General Knowledge
Indian Geography

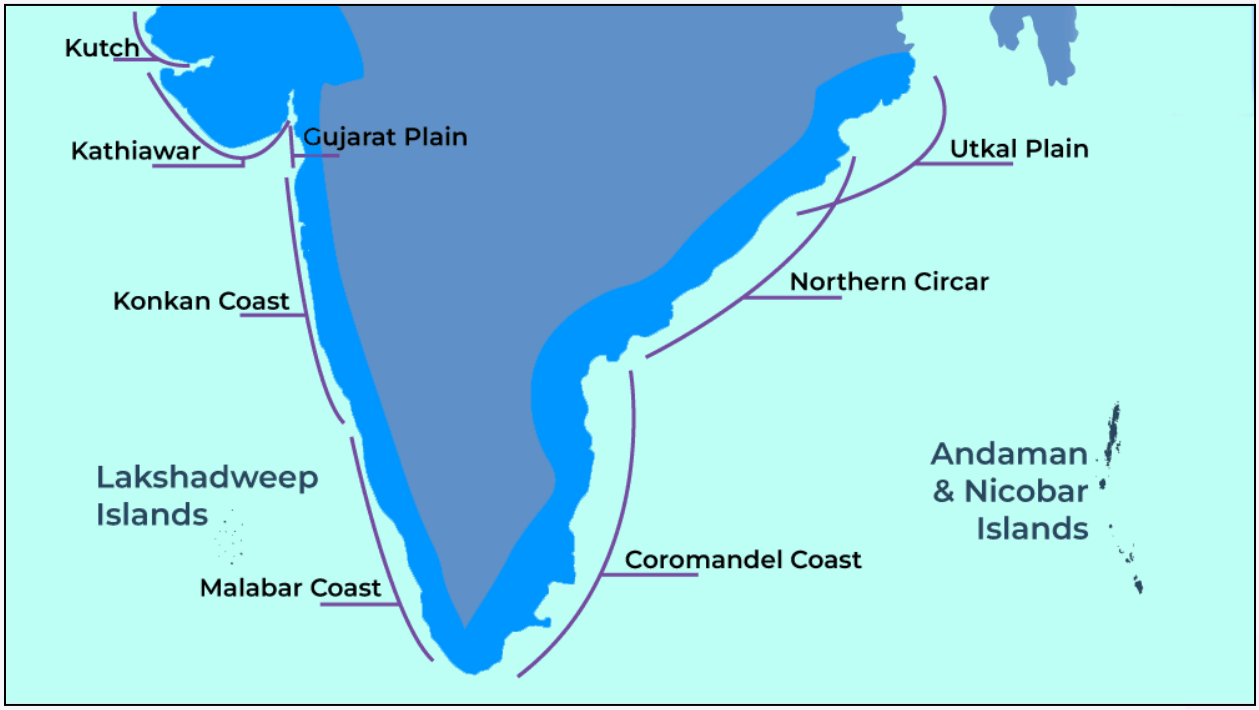
STUDY NOTES

Physiographic Divisions of India - Coastal Plains and Islands

PHYSIOGRAPHIC DIVISIONS OF INDIA - COASTAL PLAINS AND ISLANDS

The Coastal Plains of India

The **Coastal Plains of India** form a critical zone between the **Peninsular Plateau** and the coastline of India. These plains stretch along the edges of the **Arabian Sea** in the west and the **Bay of Bengal** in the east, serving as an important ecological, economic, and cultural interface between land and sea.



Key Features of Coastal Plains:

- **Total Length**: The Coastal Plains of India stretch for about **6,000 km**, extending along both the western and eastern coasts of India.
- **Geographical Position**: These plains are located between the **Peninsular Plateau** (which forms the hinterland) and the coastline of India.

- **Physiographic Division**: The Coastal Plains are one of the **five main physiographic divisions** of India.
- **Ecological Significance**: Coastal plains are home to important ecosystems, including **mangroves**, **estuaries**, and **salt marshes**, and are critical for biodiversity.
- **Economic Importance**: These areas are rich in **natural resources**, **ports**, and **agriculture**, and are central to India's **trade and commerce**.

Divisions of the Coastal Plains:

The Coastal Plains of India are divided into two major parts:

1. **Western Coastal Plains**
2. **Eastern Coastal Plains**

These two coastal sections meet at **Kanyakumari**, the southernmost tip of India.

Western Coastal Plains:

- **Location**: Between the **Western Ghats** and the **Arabian Sea** coast.
- **Spread**: From the **Rann of Kutch** in Gujarat in the north to **Kanyakumari** in the south.
- **States Covered**: **Gujarat**, **Maharashtra**, **Goa**, **Karnataka**, **Kerala**.
- **Width**: Narrower in the middle (about **65 km**) but broader in the north and south.
- **Coastline Type**: **Submergent coastline**, resulting from **eustatic sea-level rise** (submergence of land), facilitating the development of **natural harbours and ports**.

The Western Coastal Plains can be further divided based on **relief** and **topography**.

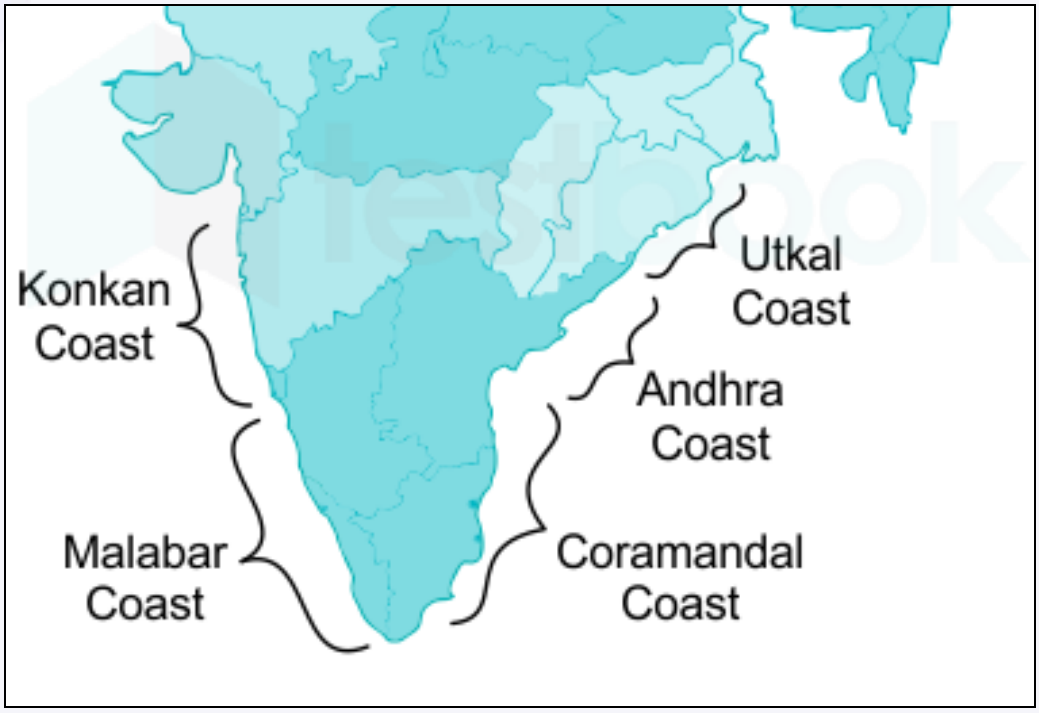
Kutch Peninsula (Northwestern Gujarat):

- **Location**: Extends over **northwestern Gujarat**.
- **Formation**: Originally an island, the Kutch Peninsula became part of the mainland due to sediment deposition by the **Indus River**.
- **Topography**: Arid and semi-arid, with features like **coastal sand dunes**, **sandy plains**, and **rocky hills**.
- **Significance**: Known for the **Great Rann of Kutch**, a salt-soaked plain and seasonal wetland.

- **Rann of Kutch**: A large **salt marsh** in the north, surrounded by salt flats and receiving little rainfall.

Konkan Coast:

- **Location**: From **Mumbai** (Maharashtra) to **Goa**.
- **Topography**: Characterized by a **narrow, rocky coastline** interspersed with **coconut groves** and small **beaches**.
- **Rivers**: The region is drained by rivers like the **Mandovi** and **Zuari**.
- **Climate**: Tropical, with high rainfall and humidity.
- **Economic Importance**: Rich in **fisheries** and has major **ports** like **Mumbai** and **Goa**.



Karnataka Coast:

- **Location**: From **Goa** to **Kerala**.
- **Features**: Known for **long beaches** like **Udupi**, **Mangalore**, and **Karwar**.
- **Rivers**: Major rivers include the **Netravati**, **Sharavati**, and **Tungabhadra**.
- **Topography**: Coastal plains here are narrow and feature **rocky promontories**, **salt pans**, and **backwaters**.

Kerala Coast:

- **Location**: The **southernmost part of the western coast**.
- **Features**: Known for **backwaters**, **lagoons**, and **beaches** like **Kochi** and **Varkala**.
- **Economic Importance**: Major agricultural activities such as **coconut farming** and **spices**. The region also has key **ports** like **Cochin Port**.
- **Rivers**: **Periyar**, **Pamba**, and **Bharathappuzha** drain the region.

Eastern Coastal Plains:

- **Location**: Located between the **Eastern Ghats** and the **Bay of Bengal** coast.
- **Spread**: From the **West Bengal** coastline in the north to **Tamil Nadu** and **Andhra Pradesh** in the south.
- **States Covered**: **West Bengal**, **Odisha**, **Andhra Pradesh**, **Tamil Nadu**.
- **Width**: Generally wider than the western counterpart, with extensive **delta regions**.
- **Coastline Type**: **Emergent coastline**, indicating the land is rising above sea level.

The Bengal Coast (West Bengal and Odisha):

- **Location**: Between **Kolkata** (West Bengal) and **Chandbali** (Odisha).
- **Features**: Characterized by the **Sundarbans mangrove forest**, the largest of its kind in the world.
- **Rivers**: Drained by the **Hooghly** and **Mahanadi** rivers.
- **Economic Importance**: Rich in **marine life**, **agriculture**, and **ports** like **Kolkata**.
- **Delta**: The **Sundarbans Delta** is an ecologically rich area, known for its **tiger reserve** and **mangrove ecosystems**.

Andhra Pradesh and Tamil Nadu Coast:

- **Location**: From **Kakinada** (Andhra Pradesh) to **Chennai** (Tamil Nadu).
- **Features**: Known for **long beaches**, **estuaries**, and **backwaters**.
- **Rivers**: Major rivers like the **Godavari**, **Krishna**, and **Cauvery** drain this region.
- **Economic Significance**: Major ports like **Chennai Port** and **Visakhapatnam** are located here, supporting India's **shipping industry**.
- **Fisheries**: The region is also a major producer of **seafood**.
- **Climate**: Tropical with seasonal monsoons.

The Coromandel Coast:

- **Location**: In **Tamil Nadu**, between **Chennai** and **Cuddalore**.
- **Features**: Known for **estuaries**, **salt marshes**, and **delta** regions.
- **Rivers**: The **Cauvery River** forms a major delta here.
- **Economic Significance**: The **Cauvery Delta** is agriculturally rich and supports **irrigated agriculture**.

Comparison Between Eastern and Western Coastal Plains

Feature	Western Coastal Plains	Eastern Coastal Plains
Width	Narrower	Wider
Geographical Location	Lies between the **Western Ghats** and the **Arabian Sea**	Lies between the **Eastern Ghats** and the **Bay of Bengal**
Topography	Intersected by various mountains of the **Western Ghats**	Runs continuously from **north to south**
Rivers	Rivers form **estuaries** (e.g., **Narmada**, **Tapi**, **Mahi**, **Godavari**, **Krishna**)	Rivers form **wide deltas** (e.g., **Mahanadi**, **Godavari**, **Krishna**, **Cauvery**)
Harbors	Favours the formation of **harbours** (e.g., **Mumbai**, **Mangalore**, **Kochi**)	Do not favour the formation of harbours due to wide continental shelves (e.g., **Kolkata** Port is an exception)
Monsoon Rains	Receives a high amount of rain from **southwest monsoons**	Receives rains from both **northeast** and **southwest monsoons**
Rainfall	High rainfall, especially along the **Konkan coast** and **Kerala**	Moderate rainfall along the **Coromandel Coast** and **Northern Andhra Pradesh**

Coastal Features	Rocky coast with **backwaters** on the **Malabar Coast** and **estuaries** on the **Konkan Coast**	Mostly **sandy** with **sand dunes**, **lagoons**, and **deltaic plains**
Fishing Industry	Rich in marine fisheries, supports **commercial fishing**	Rich in marine fisheries, with large **estuaries** ideal for fishing
Agriculture	Mostly **monsoon-dependent** agriculture (e.g., **coconut**, **rubber**, **spices**)	Agriculture is **irrigated**, focused on crops like **rice**, **sugarcane**, and **cotton**
Climate	**Tropical** climate with high humidity, moderate to heavy rainfall	**Tropical** to **subtropical** climate with moderate humidity, rainfall varies
Coastal Ecosystems	Rich in **mangrove forests**, **estuaries**, and **backwaters**	Rich in **deltaic ecosystems**, **salt marshes**, and **mangrove forests**
Major Ports	**Mumbai**, **Kochi**, **Mangalore**, **Goa**, **Kandla**, **Surat**	**Kolkata**, **Chennai**, **Visakhapatnam**, **Paradip**, **Cochin**

Significance of the Coastal Plains:

- **Economic Significance**:
  - **Ports**: The coastal plains house several major **ports** like **Mumbai**, **Chennai**, **Kochi**, and **Visakhapatnam**, facilitating **international trade**.
  - **Agriculture**: Fertile plains, especially the delta regions, support **rice**, **sugarcane**, and **fisheries**.
  - **Tourism**: The coastal areas are popular for **beaches**, **heritage sites**, and **wildlife sanctuaries**.
- **Ecological Significance**:
  - The plains are home to rich ecosystems, including **mangroves**, **salt marshes**, and **estuaries**, which provide breeding grounds for **marine life** and protect the coastline from erosion.
  - **Biodiversity**: The **Sundarbans**, **Coromandel Coast**, and **backwaters** of Kerala are rich in biodiversity and host many rare species of flora and fauna.

- **Climate Control**:
  - The coastal plains influence **monsoon patterns** and **temperature regulation**. Coastal areas tend to have a **moderate climate**, with **sea breezes** lowering the impact of extreme heat in summer.

The Islands of India

The **Indian Islands** refer to the group of islands scattered across the **Indian Ocean**, the **Arabian Sea**, and the **Bay of Bengal**. These islands form part of India's geographical territory and are an important component of its **physiographic divisions**.

There are two main groups of islands:

- **The Andaman and Nicobar Islands**
- **The Lakshadweep Islands**

Andaman and Nicobar Islands

The **Andaman and Nicobar Islands** form a significant part of India's geographical landscape, located in the **Bay of Bengal**. These islands are known for their ecological importance, strategic location, and rich biodiversity. Below is a comprehensive breakdown of the Andaman and Nicobar Islands, covering their geography, sub-groups, important islands, tribes, resources, and more.

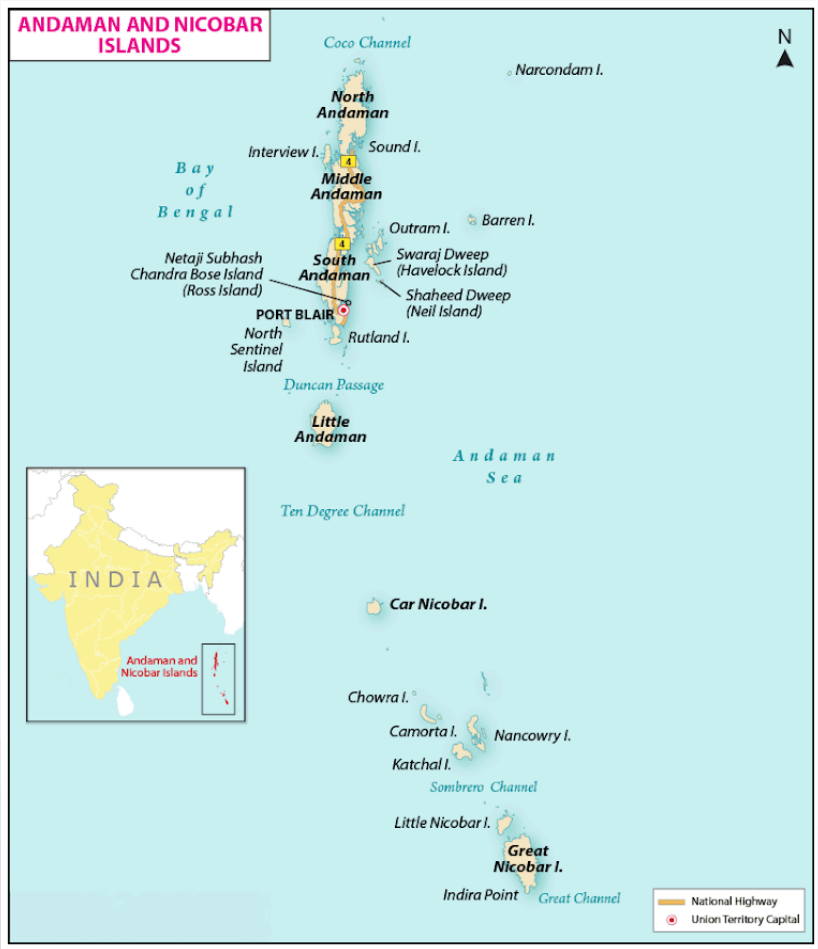
- **Location**: The islands are located in the **Bay of Bengal**, to the southeast of the Indian subcontinent, between **6° 45′ N to 13° 41′ N** latitude and **92° 12′ E to 93° 57′ E** longitude.
- **Geographical Extent**:
  - The Andaman and Nicobar Islands stretch over approximately **590 km** in length.
  - The maximum width of the archipelago is about **58 km**.
- **Administrative Division**: The islands are part of the **Union Territory of Andaman and Nicobar Islands**, with **Port Blair** as the capital.
- **Political Significance**:
  - The islands hold strategic military importance, as they are located near the **Malacca Strait**, a critical international shipping route.

Channel	Division
10-Degree Channel	The Andaman Islands and Nicobar Islands
Duncan Passage	Great Andaman and Little Andaman
St. George Channel	Little Nicobar and Great Nicobar
Grand Channel	Great Nicobar and Sumatra Islands (Indonesia)

Major Groups of Islands

The **Andaman and Nicobar Islands** are divided into two main groups:

- **The Andaman Islands**
- **The Nicobar Islands**



Each group has distinct characteristics, geography, and features.

The Andaman Islands

- **Location**: Located to the **north of the Nicobar Islands**, these islands are the first part of the archipelago encountered from the mainland of India.
- **Area**: The Andaman Islands cover approximately **2,400 sq km**.
- **Division**: The Andaman Islands are divided into three major sub-groups:
  - **North Andaman**
  - **Middle Andaman**
  - **South Andaman**
- **Key Islands**:
  - **Great Andaman**: The largest island, located in the northern part of the group.
  - **Little Andaman**: Located in the south, separated from Great Andaman by the **Duncan Passage** (50 km wide).
  - **Havelock Island**: A popular tourist destination known for its pristine beaches and water sports.
  - **Neil Island**: Known for its natural beauty and coral reefs.
- **Geographical Features**:
  - The Andaman Islands are characterized by **tropical rainforests**, **mangrove swamps**, and **coral reefs**.
  - **Volcanic activity**: The **Barren Island**, located in the northernmost part of the Andaman group, is an active volcano.
- **Climate**:
  - The region experiences a **tropical climate** with high humidity throughout the year.
  - The average annual rainfall is about **3,000 mm**.
- **Capital**: **Port Blair**, located in **South Andaman**, is the administrative capital of the Union Territory.
- **Tribes**: The Andaman Islands are home to several Indigenous tribes:
  - **Jarwas** and **Sentinalese** (isolated and hostile tribes)
  - **Onge** and **Great Andamanese** (semi-isolated tribes)
- **Biodiversity**: The islands are rich in wildlife and vegetation, with a variety of endemic species, including:
  - **Andaman wood pigeon**
  - **Andaman wild boar**

    - **Mugger crocodile**
    - **Coral reefs** around Havelock and other islands
- **Tourism**: The Andaman Islands are a popular tourist destination, with attractions like:
    - **Radhanagar Beach** (Havelock)
    - **Cellular Jail** (Port Blair)
    - **Chidiya Tapu** (bird sanctuary)

The Nicobar Islands

- **Location**: Located south of the Andaman Islands, the Nicobar Islands are the southernmost group of islands in the Indian archipelago.
- **Area**: The Nicobar Islands cover an area of about **1,653 sq km**.
- **Division**: The Nicobar Islands are divided into three major sub-groups:
    - **Northern Group**: Includes **Car Nicobar** and **Battimalv**.
    - **Central Group**: Includes **Chowra**, **Chaura**, **Teressa**, **Katchal**, **Camorta**, **Nancowry**, **Trinket**, and **Tillangchong**.
    - **Southern Group**: Includes **Great Nicobar**, **Little Nicobar**, **Kondul Island**, **Pulo Milo** (Milo Island), **Meroe**, **Treis**, and others.
- **Key Islands**:
    - **Great Nicobar**: The largest and southernmost island in the Nicobar group. It is located about **147 km** from **Sumatra** (Indonesia).
    - **Car Nicobar**: Known for the **Car Nicobar Airfield**, a strategic military base.
    - **Katchal and Camorta**: Noted for their scenic beauty and small populations.
- **Geographical Features**:
    - The Nicobar Islands are mostly **flat** with some **rolling hills** and have **dense tropical forests**.
    - The islands are located near the **Sunda Shelf**, where the ocean depth rapidly increases.
- **Climate**: The Nicobar Islands experience a **tropical climate** with seasonal rainfall, mostly from the **southwest monsoon**.
    - **The average annual rainfall** is about **3,000 mm**.
- **Tribes**: The Nicobar Islands have **Indigenous tribes** such as:
    - **Shompens** (semi-nomadic tribe)
    - **Nicobarese** (mainly settled and semi-modernized)
- **Biodiversity**: The Nicobar Islands are home to unique flora and fauna, including:

- **Nicobar pigeon** (the state bird)
- **Varieties of marine life**, including dolphins, sea turtles, and dugongs.

- **Strategic Importance**: Due to their location, the Nicobar Islands have **strategic importance** for India's defence and connectivity with Southeast Asia.

Major Rivers and Water Bodies

- **Rivers**: While there are no significant rivers flowing through the islands, several **streams** and **small rivers** flow into the sea, providing fresh water sources.
  - **The Dhanikhari River** (South Andaman) is one of the longest rivers.
- **Water Bodies**:
  - **Lakes**: Several lakes, such as **Saddle Peak Lake** (North Andaman), add to the beauty and ecological significance of the islands.
  - **Oceans and Seas**: The Andaman Sea surrounds the islands to the west, while the **Bay of Bengal** lies to the east.

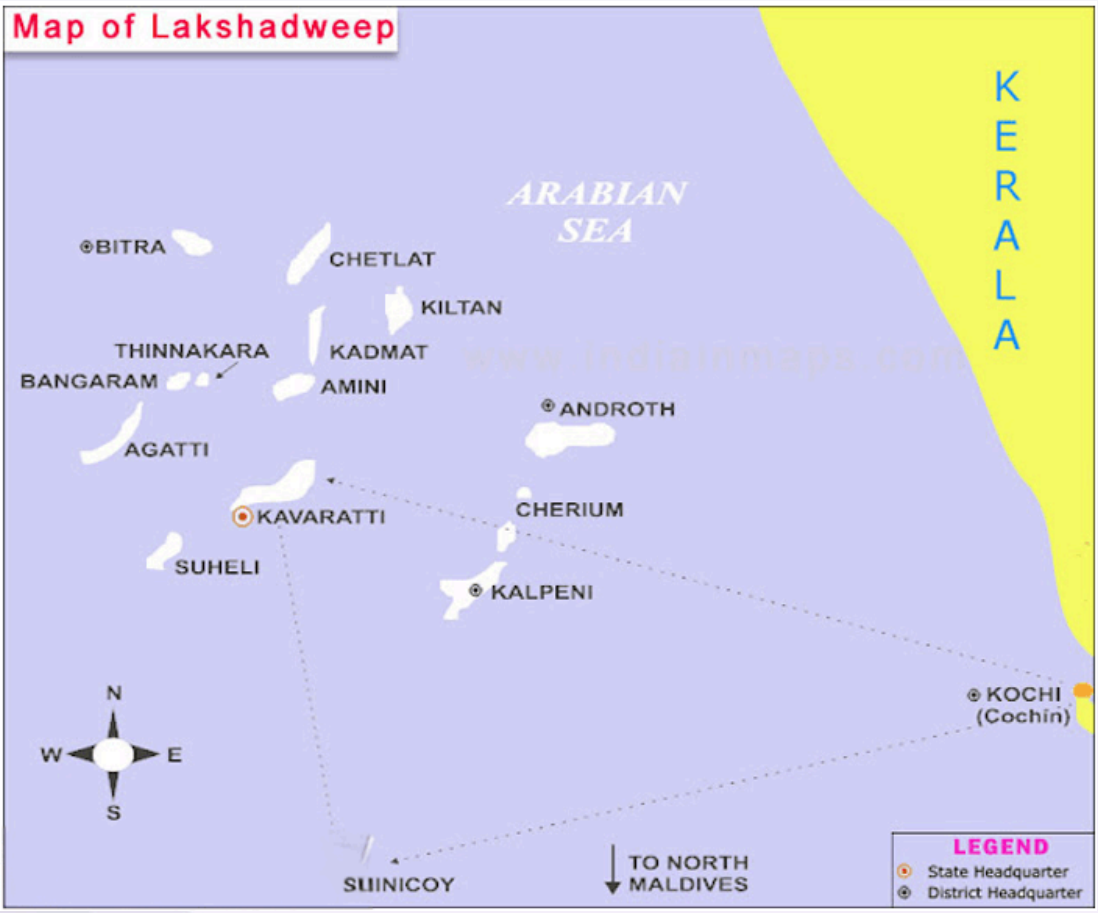
Environmental Issues

- **Deforestation**: Increased human settlement and tourism have led to deforestation in certain areas.
- **Conservation Efforts**:
  - The **Andaman and Nicobar Islands** are home to several **protected areas**, such as the **Rani Jhansi Marine National Park** and **South Reef Sanctuary**.
  - These efforts are aimed at preserving the **endangered species** and unique ecosystems of the islands.

Lakshadweep Islands

The Lakshadweep archipelago is composed entirely of **coral deposits**. All the islands are atolls surrounded by fringing reefs. These islands are located in the **Reunion Hotspot volcanic region**. The **northernmost islands** are the **Amindivi Islands**, while the **southernmost** is **Minicoy Island**. The **8°N latitude separates Minicoy Island from the Maldives**, while the **9°N latitude divides Minicoy from the rest of the Lakshadweep group**. **Minicoy Island**, located south of the 9-degree channel, is the **largest island** in the archipelago.

- **Location**: The Lakshadweep Islands are located in the **Arabian Sea**, about **200 to 400 km** off the **Kerala** coast in southwestern India. They are situated between **8° and 12° N latitude** and **71° to 74° E longitude**.
- **Number of Islands**:
  - The Lakshadweep Islands consist of **36 islands**, spread over an area of about **30,000 square miles**.
  - Out of the 36 islands, **only 10 are inhabited**.
- **Area**: The total land area of Lakshadweep is about **32.6 square km**, making it India's smallest Union Territory.
- **Capital**: The capital of Lakshadweep is **Kavaratti**, which is also one of the largest and most developed islands in the archipelago.
- **Administrative Division**: Lakshadweep is a Union Territory of India, and the **Administrator** of Lakshadweep is appointed by the President of India.



Geographic Features

- **Atolls and Coral Reefs**:
  - Lakshadweep is famous for its **coral atolls**, which are ring-shaped coral reefs that encircle lagoons.
  - The islands are **coral islands**, formed by the deposition of corals and marine organisms.
  - The **most famous atoll** in Lakshadweep is the **Agatti Atoll**.
- **Topography**:
  - The islands have **flat terrain**, and most of them are low-lying.
  - The average elevation of the islands is just **1-2 meters** above sea level.
  - These islands are formed by **coral deposits**, making the soil **sandy** and infertile.
  - Many of the islands are surrounded by **beautiful lagoons**, which add to the ecological richness of the region.
- **Rivers and Lakes**:
  - Lakshadweep does not have any significant rivers due to its low elevation and terrain. The freshwater supply primarily comes from **rainwater harvesting** and **underground water reserves**.

Major Islands of Lakshadweep

The islands in Lakshadweep are spread across the **Arabian Sea**, and they are divided into **three sub-groups**:

- **Northern Group**:
  - **Amini, Androtт, Kiltan, and Chetlat** are some of the islands in this group.
  - **Androtт Island** is the largest island in the northern group and is home to the only **mosque** in Lakshadweep, the **Jamia Masjid**.
- **Central Group**:
  - **Kavaratti Island**: The capital of Lakshadweep and one of the most developed islands.
  - **Kalapeni Island**: Known for its clear waters and coral reefs, which attract tourists.
  - **Pitti Island**: A small, uninhabited island known for its bird sanctuary.
- **Southern Group**:
  - **Minicoy Island**: The largest island in the southern group, famous for its distinct **Maldivian culture**.

- **Fisheries**: Minicoy Island is known for its **fishing industry** and is home to the **Laccadive people**.
- **The island's lighthouse**, located at the southern tip, is a famous landmark.

Climate

- **Tropical Climate**: The climate in Lakshadweep is tropical, with warm temperatures throughout the year.
- **Temperature**: Average temperature ranges between **25°C to 32°C** year-round.
- **Monsoon**:
  - The **southwest monsoon** (June to September) brings heavy rainfall to the islands.
  - The annual rainfall is around **1,200 mm to 1,500 mm**.
- **Humidity**: The islands experience high humidity levels, which are typical of tropical regions.

Biodiversity and Ecology

- **Flora**:
  - Lakshadweep has limited vegetation due to its arid soil. However, the islands are home to **coconut palms**, **breadfruit trees**, and **mango trees**.
  - **Coral reefs** are an important ecological feature, supporting a variety of marine life, including **sea turtles**, **fish species**, and **crustaceans**.
- **Fauna**:
  - The coral reefs surrounding the islands are home to a wide variety of **marine species**, including **marine turtles**, **dolphins**, **whale sharks**, and **various species of fish**.
  - **Birdlife**: The islands have a variety of seabirds, and **Pitti Island** is known for being a bird sanctuary.

Economic Activities

- **Fishing**: The primary occupation of the people of Lakshadweep is **fishing**, which is the main economic activity.
- **Coconut Cultivation**: Coconut palms are extensively cultivated and are used in various forms, including coconut oil, copra (dried coconut), and as food.
- **Tourism**:

- Tourism is a growing industry, especially eco-tourism and beach tourism.
- The islands, with their pristine beaches, coral reefs, and clear waters, attract domestic and international tourists.
- **Water sports** such as **scuba diving**, **snorkelling**, and **swimming** are popular activities for visitors.
- **Agatti Island** and **Kavaratti Island** are the most frequently visited islands by tourists.

- **Agriculture**: Aside from coconuts, a few islands also cultivate crops such as **tobacco**, **vegetables**, and **fruits** for local consumption.

Tribes and Culture

- **Inhabitants**: The population of Lakshadweep is primarily of **Indo-Aryan** descent, with a mix of **Arabian** and **Maldivian** influences.
- **Languages**: The primary language spoken is **Mahl**, which is a variety of **Dhivehi** (the language of the Maldives). Other languages spoken include **Malayalam** and **English**.
- **Religion**: The majority of the population is **Muslim**, with Islam being the predominant religion across the islands.
- **Culture**: The culture of Lakshadweep is influenced by both **Indian** and **Maldivian** traditions. The people are known for their traditional **boat-building**, **fishing**, and **dancing**.

Environmental Issues

- **Coral Bleaching**: The coral reefs are under threat from **global warming**, leading to **coral bleaching** and loss of biodiversity.
- **Climate Change**: Rising sea levels and extreme weather patterns, such as storms and cyclones, pose a threat to the low-lying islands.
- **Overfishing**: Over-exploitation of marine resources is affecting the fish populations and overall ecological balance.

Other Islands of India

Island Name	Location	State/Union Territory	Significance

New Moore Island (South Talpatti)	Bay of Bengal, Ganges-Brahma putra Delta	West Bengal/Bangladesh	A small uninhabited island emerging post-cyclone in 1970, claimed by India and Bangladesh due to potential oil and gas reserves.
Diu Island	Gulf of Cambay, Kathiawar Coast	Gujarat	Famous for the Diu Fort, beaches like Nagoa and Ghoghla, and historical Portuguese influence.
Majuli Island	Brahmaputra River, Assam	Assam	World's largest river island, a cultural and spiritual hub for Assamese neo-Vaishnavites.
Abdul Kalam Island (Wheeler Island)	Off the Odisha Coast	Odisha	India's missile testing site, crucial for defence research and technology was formerly known as Wheeler Island.
Sagar Island	Ganga Delta, Bay of Bengal	West Bengal	A prominent Hindu pilgrimage site, home to the annual Sagar Mela during Makar Sankranti.
Halliday Island	Sundarbans, Malta River	West Bengal	A wildlife refuge in the Sundarbans, known for its rich biodiversity and environmental significance.
Phumdis (Floating Islands)	Keibul Lamjao National Park, Manipur	Manipur	Floating islands in Loktak Lake, are home to the endangered Sangai deer and unique ecosystems.

Munroe Island	Confluence of Ashtamudi and Vembanadu Lakes	Kerala	An inland island at the confluence of two lakes, known for its natural beauty and cultural significance.
Kochi Islands	Arabian Sea	Kerala	A group of islands near Kochi, notable for its strategic port, history, and the famous Kochi shipyard.
Rameswaram Island	Gulf of Mannar	Tamil Nadu	Important pilgrimage site, associated with Lord Rama, and an integral part of the Palk strait between India and Sri Lanka.
Netrani Island	Arabian Sea	Karnataka	A small island near Murudeshwar, popular for its coral reefs and underwater diving.
Maharajpur Island	Krishna River, Andhra Pradesh	Andhra Pradesh	Known for its picturesque landscapes and as a part of the river islands in Andhra Pradesh.
Barren Island	Andaman Sea	Andaman and Nicobar Islands	The only active volcano in India, located in the Andaman group of islands is important for geological research.
Viper Island	Andaman Sea	Andaman and Nicobar Islands	Known for its historical significance, the site of a British colonial prison and its association with the freedom struggle.

Long Island	Andaman Sea	Andaman and Nicobar Islands	Famous for its natural beauty, and beaches and is a popular tourist destination in the Andaman archipelago.